

# AYUSH JUVEKAR

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Computer Vision / Edge AI Engineer

## PROFILE

Computer vision engineer focused on video analysis, edge inference, object detection workflows, and camera/sensor-backed research prototypes.

## EDUCATION

**Michigan Technological University**, Houghton, MI  
*M.S. Computer Science*

**CGPA: 3.9/4.0**  
Graduated Apr 24, 2026

**Pune Institute of Computer Technology**, Pune, India  
*B.E. Computer Engineering*

**GPA: 8.14/10**  
Jun 2024

## EXPERIENCE

**Freelance Software Developer** | Coherence Academy  
Pune, India

Aug 2026 - Present

- Building Coherence Fee Desk, a Next.js platform replacing the academy's spreadsheet-based admissions, fee collection, and reporting.
- Separately run the software for Coherence-backed events (Campus Connect, Mission 95): registration, UPI payments, Sheets sync.

**Teaching Assistant - Physics Department** | Michigan Technological University  
Houghton, MI

Sep 2024 - Apr 24, 2026

- Supervised and graded physics labs for 100+ students across four semesters, guiding experiments, explaining concepts, and providing clear feedback across five lab sections per semester.

## SELECTED PROJECTS

**Crowd Anomaly Detection Prototype** | *Python, TensorFlow, OpenCV, TensorFlow Lite, Jetson Nano*

- Led model research and data collection for a BE final-year team project on edge crowd-anomaly detection, recording training videos and evaluating CNN/TensorFlow approaches on 10K+ frames.
- Optimized the prototype with TensorFlow Lite for Jetson Nano demonstration, reaching about 92% test accuracy in controlled demo runs without claiming production deployment.

**Particle Vibration Analysis** | *MATLAB, Computer Vision, Object Detection, ZED 2 Camera*

- Built MATLAB computer-vision scripts for particle vibration analysis from ZED 2 camera video, using object detection workflows to track particles in test recordings.
- Delivered analysis outputs for a professor's downstream PhD research, validating scripts on sample particle videos before larger research use.

**IoT Air Quality Monitoring System** | *Arduino Uno, C++, MQTT, Zigbee, Node-RED* | [IEEE](#)

- Co-built an IEEE-published IoT air-quality prototype for CO/gas monitoring, using Arduino Uno/C++, MQTT/Zigbee communication, and Node-RED flows to automate air-filter behavior from live sensor readings.
- Owned the Node-RED automation and conference presentation for the two-person project, contributing to the paper/report accepted at ESCI 2023.

## TECHNICAL SKILLS

|                   |                                                                                    |
|-------------------|------------------------------------------------------------------------------------|
| Languages         | Python, C++, MATLAB                                                                |
| Computer Vision   | OpenCV, MATLAB Computer Vision workflows, image/video processing, object detection |
| ML / Edge AI      | TensorFlow, TensorFlow Lite, CNNs, Jetson Nano                                     |
| Sensors / Cameras | ZED 2 camera video, Arduino sensor data                                            |
| Data Workflows    | Dataset preparation, model evaluation, experiment documentation                    |
| Tools             | Jupyter Notebook, Git, Linux                                                       |

## PUBLICATION

A. Ghadge, A. Juvekar et al., "Carbon Monoxide Concentration Monitoring System for Automating Air Filters," *ESCI 2023*. [IEEE](#)